

Test of Significance :-

A very important aspect of sampling theory is the study of the test of significance; which enable us to decide on the basis of the sample results, if

(i) the deviation between the observed sample statistic and the hypothetical parameter value; or

(ii) The deviation between two independent sample statistics; is significant or might be attributed to chance or the fluctuations of sampling.

* Since for large n , all the distributions Binomial, Poisson, Chi-square etc. can be approximated very closely by a Normal probability curve; we use the Normal Test of Significance for large samples.

* Some of the well known tests of Significance for studying such differences for small samples are t-test, F-test, and Fisher's Z-transformation.

Statistical Hypothesis

In attempting to arrive at decision about the population on the basis of sample information it is necessary to make assumption about population parameters such an assumption is called hypothesis which may or may not be true.

Any statement about population parameter is called hypothesis.

Hypothesis can be divided into two parts:

Null Hypothesis: Any statement about population parameter which is assumed to be true and is tried for its possible rejection is called null hypothesis. As the name suggests, it is always taken as a hypothesis of no difference usually denoted by H_0 .

Alternative Hypothesis: Any statement which is complementary to the null hypothesis is called alternative hypothesis usually denoted by H_1 . It is very important to state the alternative hypothesis in respect to any null hypothesis because the acceptance or rejection of H_0 is meaningful only if it is tested against a rival hypothesis.

explaining things in the context of “The Court Case”?

Members of a jury have to decide whether a person is guilty or innocent based on evidence

Null: The person is innocent

Alternative: The person is not innocent (i.e. guilty)

The null can only be rejected if there is enough evidence to doubt it

i.e. the jury can only convict if there is beyond reasonable doubt for the null of innocence

They do not know whether the person is really guilty or innocent so they may make a mistake

Examples

For Example, one may wish to test

$$H_0 : \mu = \mu_0 = 100 \quad (\text{mean I.Q. of a Population})$$

versus / against

$$H_1 \text{ or } H_A : \mu \neq \mu_0 = 100 \quad (\text{Two sided alternative})$$

OR (Two tailed alternative)

$$H_1 \text{ or } H_A : \mu > \mu_0 = 100 \quad (\text{One sided alternative})$$

OR (Right tailed)

$$H_1 \text{ or } H_A : \mu < \mu_0 = 100 \quad (\text{One sided alternative})$$

(Left tailed)

Example:

- Suppose a new type of blood test is introduced.
- A patient has some disease and he wants to get it tested with the blood test.
- We have following four possibilities.

		Test	
Disease		Detects	Not detects
	Presence	Correct decision	Incorrect decision
	Absent	Incorrect decision	Correct decision

Error 1: Person has disease and test does not detects.

Error 2: Person does not has disease and test detects it.

Errors in Sampling :-

The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In practice we decide to accept or reject the lot after examining a sample from it. As such we are liable to commit the two types of errors:-

Type I Error

A type I error is also known as a false positive and occurs when a researcher incorrectly rejects a true null hypothesis. This means that your report that your findings are significant when in fact they have occurred by chance.

Type I Error :- Reject H_0 when it is true.

Type II Error

A type II error is also known as a false negative and occurs when a researcher fails to reject a null hypothesis which is really false. Here a researcher concludes there is not a significant effect, when actually there really is.

Type II Error :- Accept H_0 when it is wrong.
i.e. Accept H_0 when H_1 is true.

Errors of type I and type II

α = Probability of type I error

= Probability of rejecting H_0 when H_0 is true

β = Probability of type II error

= Probability of accepting H_0 when H_0 is false.

Symbolically

$$P(x \in W | H_0) = \alpha \Rightarrow$$

$$\int_W L_0 dx = \alpha$$

where L_0 is the likelihood fn. of the sample observations under H_0 .

and $\int dx$ represents the n -fold integral $\int \dots \int dx_1 dx_2 \dots dx_n$

$$\text{Again } P(x \in \bar{W} | H_1) = \beta = \int_{\bar{W}} L_1 dx = \beta$$

where L_1 is the likelihood function of the sample observations under H_1 . Since

$$\int_W L_1 dx + \int_{\bar{W}} L_1 dx = 1$$

$$\Rightarrow \int_W L_1 dx = 1 - \int_{\bar{W}} L_1 dx = 1 - \beta$$

$$\Rightarrow \boxed{P(x \in W | H_1) = 1 - \beta}$$

The value of the power function at a parameter point is called power of the test.

$\Rightarrow (1 - \beta)$ is called the power function of the test Hypothesis H_0 against alternative Hypo. H_1 .

Q. Given the frequency function:-

$$f(x, \theta) = \begin{cases} \frac{1}{\theta} & ; 0 \leq x \leq \theta \\ 0 & ; \text{elsewhere} \end{cases}$$

and that you are testing the null hypothesis $H_0: \theta = 1$ against $H_1: \theta = 2$, by means of a single observed value x . What would be the sizes of the type I and type II errors; If you choose the interval
(i) $0.5 \leq x$, (ii) $1 \leq x \leq 1.5$ as the critical region?
Also obtain the power function of the test.

Solⁿ. Here we want to test $H_0: \theta = 1$,
Against $H_1: \theta = 2$.

(i) Here $W = \{x: 0.5 \leq x\} = \{x: x \geq 0.5\}$
and $\bar{W} = \{x: x \leq 0.5\}$
 $\alpha = P\{x \in W \mid H_0\} = P\{x \geq 0.5 \mid \theta = 1\}$
 $= P\{0.5 \leq x \leq \theta \mid \theta = 1\}$
 $= P\{0.5 \leq x \leq 1\} = \int_{0.5}^1 [f(x, \theta)]_{\theta=1} dx$
 $= \int_{0.5}^1 1 dx = 0.5$

[α = Probability of type 1 error]

Similarly β = Probability of type II error

$$= P \{ x \in \bar{W} / H_1 \} = P \{ x \leq 0.5 \mid \theta = 2 \}$$

$$= \int_0^{0.5} [f(x, \theta)]_{\theta=2} = \int_0^{0.5} \frac{1}{2} dx = 0.25$$

Thus the sizes of type I and type II errors are respectively $\alpha = 0.5$ and $\beta = 0.25$.

and Power function of test $= 1 - \beta$
 $= 1 - 0.25$
 $= 0.75.$

$$(ii) \quad W = \{ 1 \leq x \leq 1.5 \}$$

$$\alpha = P \{ x \in W / \theta = 1 \} = \int_1^{1.5} \{ f(x, \theta) \}_{\theta=1} dx = 0$$

Since under $H_0 : \theta = 1$, $f(x, \theta) = 0$ for $1 \leq x \leq 1.5$

$$\beta = P \{ x \in \bar{W} / \theta = 2 \} = 1 - P \{ x \in W / \theta = 2 \}$$

$$= 1 - \int_1^{1.5} \{ f(x, \theta) \}_{\theta=2} dx = 1 - \left(\frac{x}{2} \right)_1^{1.5} = 0.75$$

Power function $1 - \beta = 1 - 0.75 = 0.25$.

Q. If $x \geq 1$ is the critical region for testing $H_0: \theta = 2$ against the alternative $\theta = 1$, on the basis of the single observation from the population
$$f(x, \theta) = \theta \exp(-\theta x); \quad 0 \leq x < \infty$$

obtain the values of type I and type II error.

Solⁿ. Here $W = \{x: x \geq 1\}$ and $\bar{W} = \{x: x < 1\}$

and $H_0: \theta = 2; H_1: \theta = 1$.

α = Size of Type I error

$$= P\{x \in W | H_0\} = P\{x \geq 1 | \theta = 2\}$$

$$\alpha = \int_1^{\infty} [f(x, \theta)]_{\theta=2} dx = 2 \int_1^{\infty} e^{-2x} dx = 2 \left(\frac{e^{-2x}}{-2} \right)_1^{\infty}$$

$$\boxed{\alpha = \frac{1}{e^2}}$$

$\beta =$ Size of type II error

$$= P[X \in \bar{W} | H_1] = P\{X < 1 | \theta = 1\}$$

$$= \int_0^1 e^{-x} dx = \left(\frac{e^{-x}}{-1} \right)_0^1 = \boxed{\frac{e-1}{e} = \beta}$$

Level of Significance

This refers to the degree of significance with which we accept or reject a particular hypothesis. Since 100% accuracy is not possible in taking a decision over the acceptance or rejection of a hypothesis, we have to take a decision at particular level of confidence which would speak of the probability of one being correct or wrong in accepting rejecting a hypothesis.

Critical Region

A *critical region*, also known as the rejection region, is a set of values for the test statistic for which the null hypothesis is rejected. i.e. if the observed test statistic is in the critical region then we reject the null hypothesis and accept the alternative hypothesis.

A region (corresponding to a statistic t) in the sample space S which amounts to rejection of H_0 is termed as critical region of rejection.

If ω is critical region and if $t = t(x_1, x_2, \dots, x_n)$ is the value of statistic based on a random sample of size n ; then

$$P(t \in \omega | H_0) = \alpha, \quad P(t \in \bar{\omega} | H_1) = 1 - \beta$$

Where $\bar{\omega}$, the complementary set of ω , is called acceptance region.

We have $\omega \cup \bar{\omega} = S$ and $\omega \cap \bar{\omega} = \phi$

Level of Significance :- The probability ' α ' that a random value of the statistic t belongs to the critical region is known as the level of significance. In other words, level of significance is the size of the type - I error.

- * the level of significance usually employed in testing of hypothesis are 5% and 1%.
- * The level of significance is always fixed in advance before collecting the sample information.

One-tailed and Two-tailed Tests:-

* In any test the critical region is represented by a portion of the area under the probability curve of the sampling distribution of the test statistic.

① one tailed Test :- A test of any statistical hypothesis where the alternative hypothesis is one tailed (Right tailed or Left tailed) is called one-tailed test.

* For example : A test for testing the mean of a population $H_0 : \mu = \mu_0$ against the alternative Hypothesis :-

* In the right tailed test ($H_1: \mu > \mu_0$) the critical region lies entirely in the right tail of the sampling distribution of $\bar{x}$.

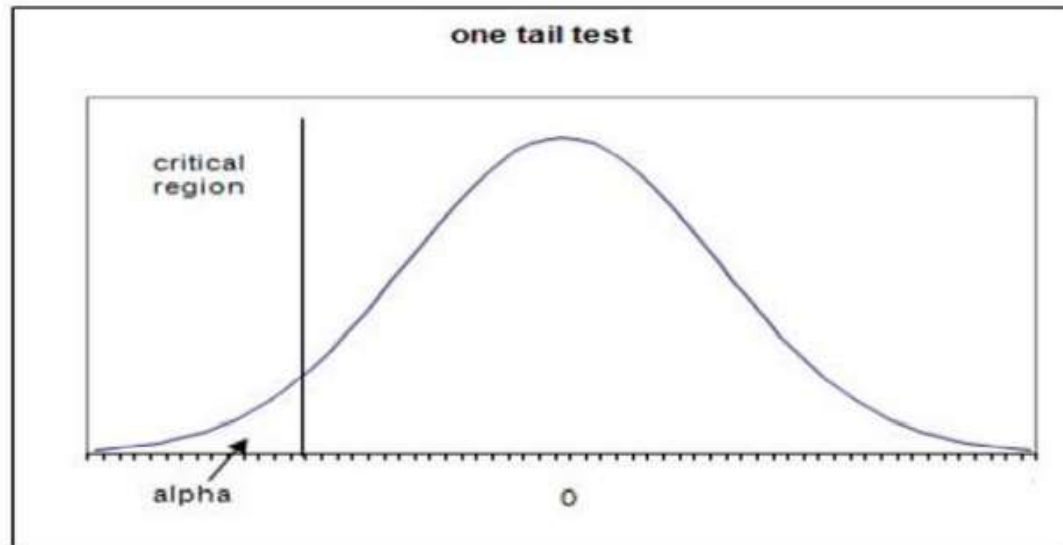
* While for the left tailed test ($H_1: \mu < \mu_0$) the critical region is entirely in the left tail of the distribution.

- One tail tests:

$$H_0: \mu = \mu_0$$

$$H_1: \mu < \mu_0;$$

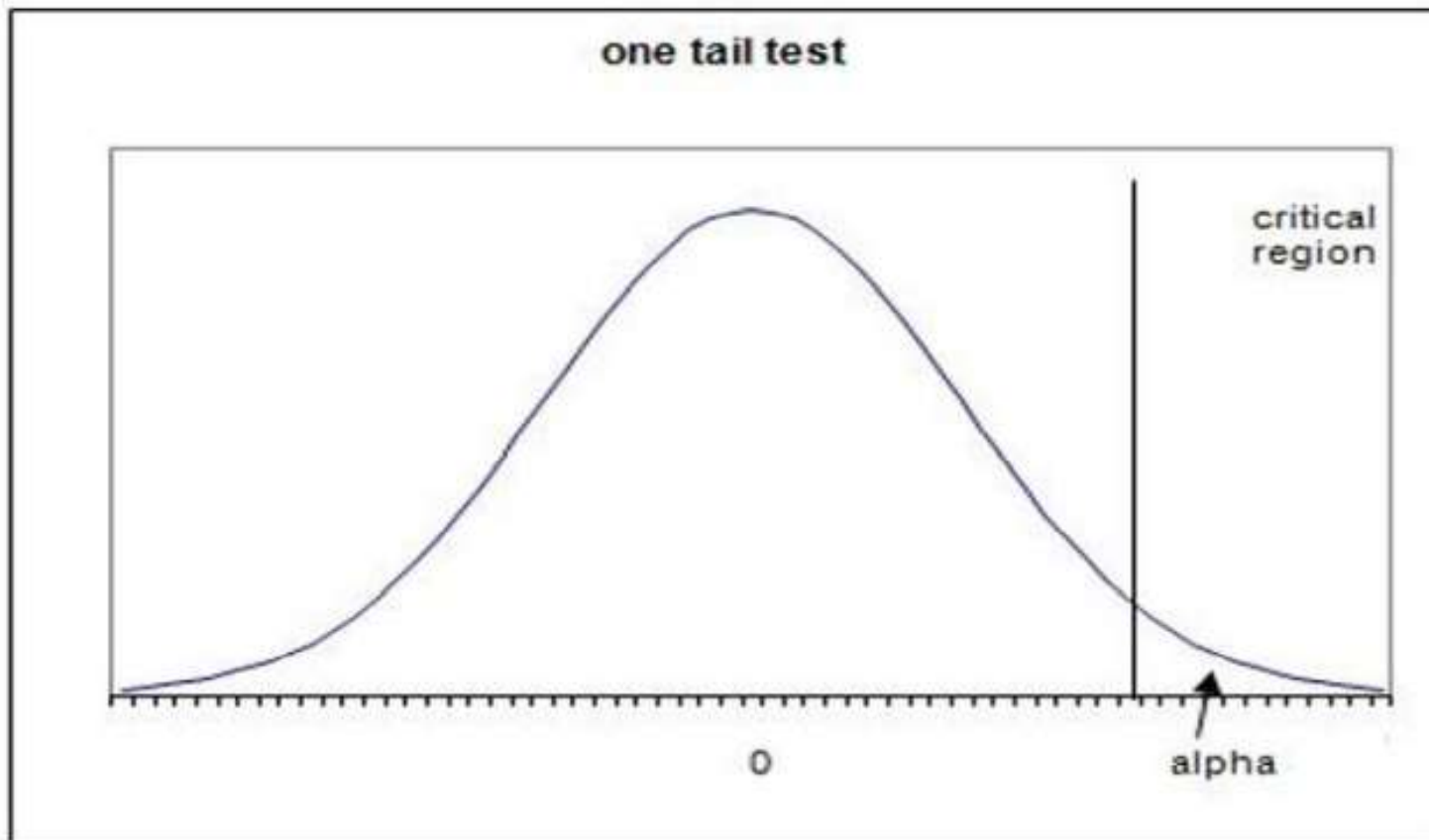
(left tailed)



$$H_0 : \mu = \mu_0$$

$$H_1 : \mu > \mu_0;$$

(Right tailed)



1. One tailed and two tailed tests

The statistical tests used will be **one tailed** or **two tailed** depending on the nature of the null hypothesis and the alternative hypothesis.

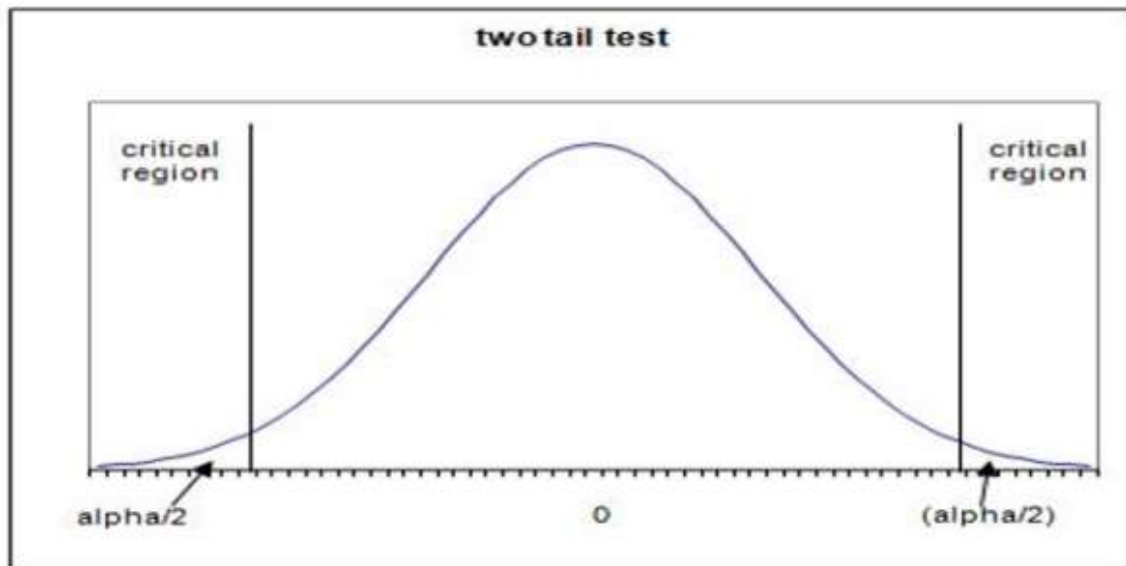
The following hypothesis applies to test for the mean:

- **two tailed test:**

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0;$$

$$[\mu > \mu_0 \text{ and } \mu < \mu_0]$$



[in such cases the critical region is given by the portion of the area lying in both tails of the probability curve of the test statistic.]

Remarks

① In a particular problem, whether one-tailed or two-tailed test is to be applied depends entirely on the nature of the alternative hypothesis

② If the alternative hypothesis is two-tailed, we apply two-tailed test and if alternative hypothesis is one-tailed; we apply one-tailed test.

Critical Values

The *critical value* at a certain significance level can be thought of as a cut-off point. If a test statistic on one side of the critical value results in accepting the null hypothesis, a test statistic on the other side will result in rejecting the null hypothesis.

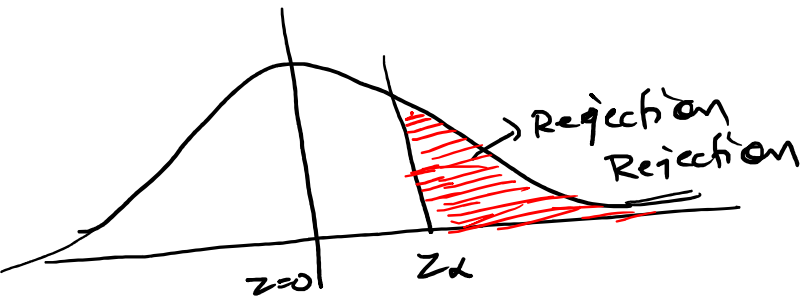
Two tailed Test (level of significance ' α ')



$$Z = \frac{x - \mu}{\sigma}$$

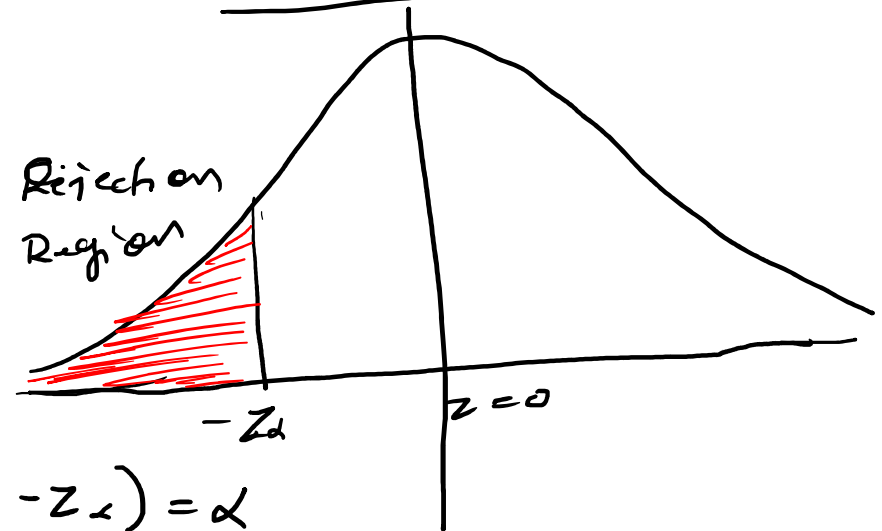
* $\rightarrow$ Significant or critical value of Z for a single-tailed test (left/right) at level of significance ' α ' is same as the critical value of Z for a two-tailed test at level of significance ' 2α '.

Right-tailed test



$$P(Z > z_\alpha) = \alpha$$

Left-tailed test



$$P(Z < -z_\alpha) = \alpha$$

Test Statistic

It is a statistic which is used to decide whether to reject or accept H_0 . The formula for computing the value of test statistics depends upon the parameter we are testing.